

Paper: Efficient Retrieval Augmentation for Desktop Search

Authors: Internal Research Group

Abstract

We evaluate retrieval quality under constrained local compute.

The study compares lexical retrieval, semantic retrieval, and hybrid scoring.

Method

A controlled corpus with deterministic content is indexed in SQLite FTS5.

Semantic vectors are generated with quantized embedding models.

Results

Hybrid retrieval outperforms lexical-only systems on conceptual tasks.

Latency remains acceptable with bounded candidate expansion.

Conclusion

High-precision lexical matching should be preserved.

Semantic scoring should augment, not replace, deterministic ranking.

note line 1